$$\sum_j \delta_{ij} F_j = F_i.$$

That's because the only nonzero term in the sum is the one where $j = i$. Within the summation, the Kronecker symbol filters out all the F's except F_i. The obvious generalization is to define a new function that has similar filtering properties when used inside an integral. In other words, we want a new entity

$$\delta(x - x')$$

with the property that, for any function F (x),

$$\int_{-\infty}^{\infty} \delta(x - x')\ F(x')dx' \ = \ F(x). \qquad (8.4)$$

Eq. 8.4 defines this new entity, called the *Dirac delta function*, which turns out to be an essential tool in quantum mechanics. But despite its name, it isn't really a function in the usual sense. It is zero whenever $x \neq x'$, but when $x = x'$ it is infinite. In fact it is just infinite enough that the area under $\delta(x)$ equals 1. Roughly speaking, it is a function that is nonzero over an infinitesimal interval ε, but on that interval it has the value $1/\varepsilon$. Thus, its area is 1, and, more importantly, it satisfies Eq. 8.4. The function

$$\frac{n}{\sqrt{\pi}} e^{-(nx)^2}$$

approximates the delta function reasonably well as n becomes very large. Fig. 8.1 plots this approximation for increasing values of n. Even though we stop at $n = 10$, a very small value, notice that the graph has already become very narrow and sharply peaked.